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Gaia VR: A Virtual Reality-based Neuro and Biofeedback Meditation Application

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***Abstract*—Meditation and mindfulness are practices that can reduce psychological stress and stress-related health problems during times of crisis such as the current global pandemic caused by COVID-19. Promoting well-being at all ages is essential to sustainable development, and in this study, we proposed a neuro and biofeedback meditation virtual reality application and heart rate band called Gaia VR to make meditation fun, measurable, and personalized. This system could also potentially reduce the stress people have been experiencing in the pandemic. By implementing data such as heart rate variability and Alpha brainwaves, users can get feedback on their daily stress level and interact with the virtual reality environment through their meditation based on how relaxed they are through the process.**

***Index Terms*—Mindfulness, gamification, educational games, meditation, MBSR, EEG, electroencephalography, VR, Virtual Reality**

I. Introduction

An estimated 284 million people experienced an anxiety disorder among the world in 2017, making it the most prevalent mental health or neurodevelopmental disorder [1]. More and more people are suffering from psychological stress and stress-related health problems due to COVID-19, and mindfulness meditation has been mentioned in many studies that it can reduce stress and anxiety and improve mental well-being [2]. However, many people find meditation either hard to establish as a habit or have trouble learning the ”right” way or feel that they cannot experience the benefits [3]. To solve these problems, we created a neuro and biofeedback meditation virtual reality (VR) application (Gaia VR) and a heart rate band (Gaia Band) to make meditation fun, measurable, and personalized. We want to promote relaxation and bring a piece of mind to our users with our wearable and VR solution, especially those who might not have a lot of access to mental health resources.

II. Background

The applications of technology to mindfulness practices are extensive and have shown effectiveness in the recent years. We have focused on noninvasive psychophysiological measurements such as heart rate variability (HRV) and electroencephalograms (EEG). We monitored and visualized the biofeedback during the meditation, since both data have been proven to help with stress management [4] [5].

Our solution consists of two parts. First, we built a heart rate band to measure and calculate the HRV, which indicates the user's stress level. According to research [6], HRV directly reflects the health of one's nervous system. Improving one's HRV means improving their ability to adapt to change and stress, whereas a declining HRV is an indication that they are burning out.

Secondly, an EEG headband was used to monitor the brainwave signal, and in our case, we were mostly interested in alpha brainwaves because it reflects whether the user is relaxed or not. Alpha waves are often associated with a relaxed, calm and lucid state of mind, and they can be found in the occipital and posterior regions of the brain [7]. In most adults, alpha waves range in frequency from 9 to 11 Hz. User's interaction in the VR meditation is based on users' relative alpha brain power recorded via the EEG headband.

III. Gaia VR

A. System Architecture

As mentioned, our solution consists a heart rate band and an EEG device. In our prototype, the alpha brain-

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wave was what was collected and the data that reflected whether the user is relaxed or not. The user's interaction in the meditation VR is based on users' relative alpha brainwave recorded through a Muse 2 headband.

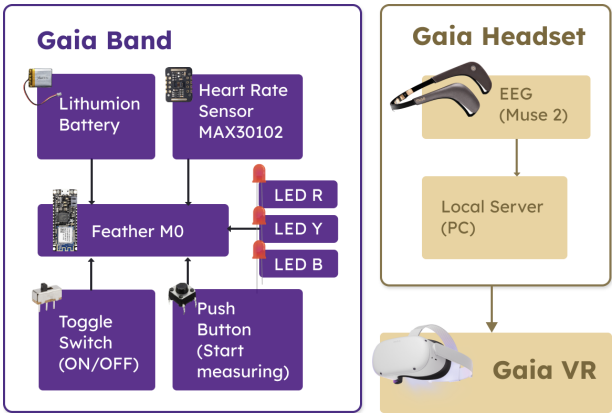

Fig. 1. Hardware Diagram

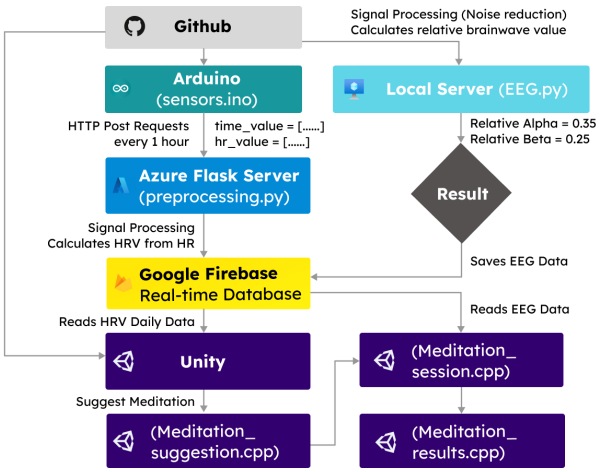

Fig. 2. Software Diagram

B. Hardware

a) Gaia Band: To obtain the user's heart rate in PPG value, we designed Gaia Band, a band-type data measuring instrument. We chose MAX30102, a PPG sensor, as the core sensor of the Gaia Band (Fig. 3). The sensor is located at the bottom of the band and measures the user's data on the wrist just like the commercially available smart bands. The data is collected through a microprocessor, Feather M0 which includes ESP8266 wifi module, and a 350mA lithium-ion battery is installed for wireless operation. For the convenience of development and debugging, an on/off slide switch, a data measurement start button, and three LEDs for checking status for network connection, measurement, and network communication were placed at the top. All components were soldered.

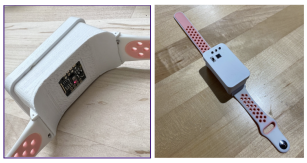

Fig. 3. Gaia Band

b) VR Solution: We used Muse 2 (Fig. 4), an EEG sensor device, to obtain the raw data. Muse 2 was placed on a participant's head, then the electrodes non-invasively detected brainwaves from the subject. The reason why Muse2 was chosen as an EEG data collection tool is that it is a device that is actively used in several studies because it provides reliable data. For the VR device, we chose Meta Quest 2. Meta Quest 2 is a standalone VR device that has the advantage that it can be used anywhere at a relatively low price compared to competitive products. Since Muse 2 and VR devices were worn on similar body parts, the two devices may interfere with each other's wear. After some usability testing, we found that there were no issues using both devices at the same time. A flexible combination of the two devices could be made possible by adding an EEG sensor to the Meta Quest 2's detachable facial interface.

Fig. 4. Muse 2

C. Software Development

a) HRV and Stress Level: To measure users' stress level, we collected the photoplethysmography(PPG) data through the band's firmware developed in the Arduino environment. The collected PPG data is then sent to the Azure Flask server and converted into HRV values, represented by the SDNN variable [8], by a program we developed using HeartPy, a Python library [9]. We then classified four stress levels:low, medium, high, and very high, based on the HRV Value according to the correlation discovered by previous studies [10].

b) EEG data: EEG data collection for Muse 2 was performed using the open source library from Petal Metrics. We processed the EEG data with a Python program, which used Fast-Fourier Transform to identify the

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average absolute Alpha wave value and then calculated its percentage in the total brainwave values [11]. All processed data is stored in a database of Google Firebase.

c) Gaia VR: The Gaia VR application was developed using Unity and the assets used in the application were free assets provided by the Unity Asset Store. It runs exclusively on Meta Quest 2, and the application can provide interactive feedback to participants by reading values from a database in real time.

D. Application Content

The user journey of Gaia VR is shown in Fig.5. During the day, the user wears Gaia Band to detect the heart beat and obtain the HRV, which indicates the stress level. According to the stress level, Gaia VR suggest specific meditation methods. Wearing the VR headset and Muse 2, the user can trigger VR interactions, such as blossoms and animal, with their brainwave during meditations. In the end, Gaia VR displays the user's achievements, including the duration of ideal brain state, to encourage a long term habit of meditation. The users can continuously build various worlds in different meditation sessions and Gaia VR provides multiple themes to create a rich experience.

Fig. 5. User Journey of Gaia VR

IV. Usability Experiments

a) Validation of the HRV measurement: For the evaluation of our HRV signal processing, we recruited 5 participants, and measured their heart rate with three different devices for 1 continuous minute. Then we calculated and recorded the average heart rate for further analysis.

Our evaluation on the calculated HRV consisted of two parts. First, we evaluated the beat per minute (BPM) data, which was calculated based on the heart rate data collected through our sensors and circuits. We chose the Finger Pulse Oximeter (FPO) data as the ground truth as it has been widely recognized as a reliable device to measure heart rate in medical applications.

Apart from Gaia Band and FPO, we also validated the measurement from Apple Watch, of which the data we used as ground truth in the evaluation of our calculated HRV. Given the result data, we applied two linear regressions, taking the Gaia Band data "Y_1" and the Apple Watch data "Y_2" as dependent variables, and the FPO data as the independent variable X. The reliability of the two evaluated data could be represented by the slope and the "R^2".

We found the equation between the BPM measured by Gaia Band and the ground truth:

$$Y_1 = 0.939 * X + 5.13, R^2 = 0.988 \quad (1)$$

We also found the equation between the BPM measured by Apple Watch and the ground truth:

$$Y_2 = X + 0.603, R^2 = 0.99 \quad (2)$$

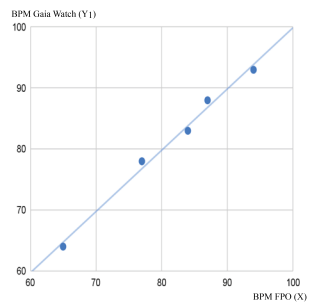

Fig. 6. BPM Gaia & BPM FPO
Linear regression between the BPM measured by Gaia Band and the BPM measured by FPO

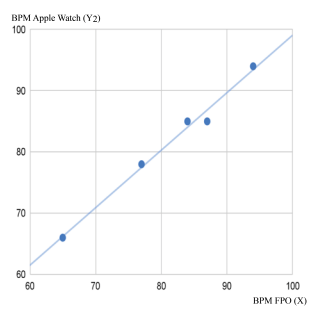

Fig. 7. BPM Apple & BPM FPO
Linear regression between the BPM measured by Apple Watch and the BPM measured by FPO

The slope represents how close the measured data is to the ground truth. The R squared value represents the goodness of fit. In the most ideal case, both the slope and the R squared value equals 1. Therefore, we found a strong correlation between both the Gaia Band data and the Apple Watch data with the ground truth, indicating

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that both devices are reliable in terms of measuring heart rate(1)(2).

Second, we evaluated the HRV from Gaia Band with the data from Apple Watch as the ground truth. We applied another linear regression, with Gaia Band HRV as the dependent variable "Y_3" and the Apple Watch HRV as the independent variable "Z". We found the following correlation:

$$Y_3 = 0.87 * Z + 6.05, R^2 = 0.976 \quad (3)$$

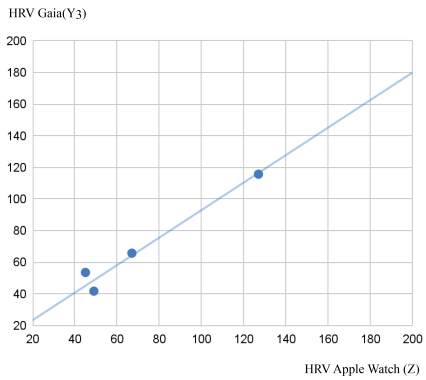

Fig. 8. HRV Gaia & HRV Apple Watch
Linear regression between the HRV calculated by Gaia and the HRV calculated by Apple Watch

We concluded that the HRV calculated by our Gaia Band program is strongly correlated to that of Apple Watch(3).

b) EEG measurement: Limitations and Future Work: In this study, we did not validate the EEG data due to the following reasons. First, the Gaia Band was our original design of sensors and circuits, while the EEG data was collected through Muse 2, a commodity device. Given limited time, we found it relatively more necessary to provide a convincing evaluation of the HRV data instead of the EEG data. Secondly, we lacked a more advanced EEG device to get ground truth data for the EEG evaluation. We could conduct more rounds of evaluation of the EEG data if we have access to a gel-based EEG device for medical level experiments in the future.

V. Discussion

In this paper, we briefly described how we constructed a virtual reality meditation application and wearable that can sense users' health data so the system can provide personalized meditation suggestions daily. In the future, we would continue to work on the contents and usability testing of the system. We could also do a longitudinal study on whether the VR meditation application could really help reduce stress in a long term, and identify other potential use cases for future studies.

VI. Limitations

We view this project as a proof of concept of a novel meditation application combining neuro and biofeedback with VR technology. Therefore, we mainly focused on the concept design and implementation of the application. For future works, we could conduct a rigorous evaluation on the consequences of our intervention. Given the limited time and budget, we validated the measured HRV values with a limited amount of recruited participants. For our next steps, we would expand the size of participants for a more convincing validation of the measured values.

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References

[1] H. Ritchie and M. Roser, "Mental health. our world in data," *Retrieved October*, vol. 7, p. 2019, 2018.

[2] J. Diemer, A. Mühlberger, P. Pauli, and P. Zwanzger, "Virtual reality exposure in anxiety disorders: Impact on psychophysiological reactivity," *The World Journal of Biological Psychiatry*, vol. 15, no. 6, pp. 427–442, 2014.

[3] E. Shapiro and D. Shapiro, "Reasons why meditation appears so difficult," *Article.(20 May 2012). Retrieved July*, vol. 20, p. 2016, 6.

[4] A. Conrad, A. Müller, S. Doberenz, S. Kim, A. E. Meuret, E. Wollburg, and W. T. Roth, "Psychophysiological effects of breathing instructions for stress management," *Applied psychophysiology and biofeedback*, vol. 32, no. 2, pp. 89–98, 2007.

[5] T. Pereira, P. R. Almeida, J. P. Cunha, and A. Aguiar, "Heart rate variability metrics for fine-grained stress level assessment," *Computer methods and programs in biomedicine*, vol. 148, pp. 71–80, 2017.

[6] I. Cygankiewicz and W. Zareba, *Autonomic Nervous System: Chapter 31. Heart rate variability*. Elsevier Inc. Chapters, 2013, vol. 117.

[7] B. Stinson and D. Arthur, "A novel eeg for alpha brain state training, neurobiofeedback and behavior change," *Complementary therapies in clinical practice*, vol. 19, no. 3, pp. 114–118, 2013.

[8] W. Wu, Y. Gil, and J. Lee, "Combination of wearable multi-biosensor platform and resonance frequency training for stress management of the unemployed population," *Sensors*, vol. 12, no. 10, pp. 13 225–13 248, 2012.

[9] P. van Gent, H. Farah, N. van Nes, and B. van Arem, "Heartpy: A novel heart rate algorithm for the analysis of noisy signals," *Transportation Research Part F: Traffic Psychology and Behaviour*, vol. 66, pp. 368–378, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1369847818306740

[10] N. Munla, M. Khalil, A. Shahin, and A. Mourad, "Driver stress level detection using hrv analysis," in *2015 International Conference on Advances in Biomedical Engineering (ICABME)*, 2015, pp. 61–64.

[11] P. Welch, "The use of fast fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms," *IEEE Transactions on audio and electroacoustics*, vol. 15, no. 2, pp. 70–73, 1967.

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